PID/DATA MONITOR INSPECTION [i-ELOOP]

id130201000500

1. Connect the M-MDS to the DLC-2.
2. After the vehicle is identified, select the following items from the initialization screen of the M-MDS.
   (1) Select “DataLogger”.
   (2) Select “Modules”.
   (3) Select “DCDC”.
3. Select the applicable PID from the PID table.
4. Verify the PID data according to the directions on the screen.

Note

- The PID data screen function is used for monitoring the calculated value of input/output signals in the module. Therefore, if the monitored value of the output component is not within the specification, it is necessary to inspect the monitored value of input component corresponding to the applicable output part control. In addition, because the system does not display an output part malfunction as an abnormality in the monitored value, it is necessary to inspect the output component individually.
- When detecting DTCs, PIDs related to a malfunctioning system may not display even if the module is normal. Therefore, if a PID is not displayed, it is necessary to verify the DTC, perform malfunction diagnosis of the DTC that was detected, and do repairs.
- The [Type] in the PID/data monitor table is indicated as follows.
  — Input: Value input to module
  — Conversion: Value converted into value input to module
  — Calculation: Value calculated for several input values in module
  — Output: Value output by module

—: Not applicable

Item	Unit/Condition	Type	description	Condition/Specification (Reference)
ALT_STOP_REQ	On/Off	Output	Displays the generator output stop request send status. The DC-DC converter (i-ELOOP) sends the generator output stop request to the PCM via CAN communication.	• On: Generator output stop request is sent • Off: Generator output stop request is not sent
BYPS_RLY	Open/Close	Input	Displays the bypass relay connection request reception status. The DC-DC convertor (i-ELOOP) receives a bypass relay connection request from the PCM via CAN communication.	• Open: Bypass relay connection request is received • Close: Bypass relay connection is not received
BYPS_RLY_TEMP	°C, °F	Conversion	Displays the bypass relay temperature.	• Displays the bypass relay temperature.
BYPS_STAT	Bypass Off/Bypass On/Bypass Off fault/ Bypass On fault	Output	Displays the bypass relay status.	• Bypass OFF: Bypass relay OFF • Bypass On: Bypass relay ON • Bypass Off fault: Bypass relay OFF (Fail) • Bypass On fault: Bypass relay ON (Fail)
CAP_DIS_RLY	Connect (On)/ Disconnect (Off)/ Fault (On)/Fault (Off)	Output	Displays the capacitor (i-ELOOP) relay status.	• Connect (On): Capacitor (i-ELOOP) relay ON • Disconnect (Off): Capacitor (i-ELOOP) relay OFF • Fault (On): Capacitor (i-ELOOP) relay ON (Fail) • Fault (Off): Capacitor (i-ELOOP) relay OFF (Fail)
CAP_SYS10	On/Off	Output	Displays the open ciucuit condition of the capacitor circuit.	• On: Capacitor (i-ELOOP) open cuircuit • Off: Capacitor circuit is normal

Item	Unit/Condition	Type	description	Condition/Specification (Reference)
CAP_SYS 14	On/Off	Output	Displays the deterioration condition of the capacitor capacitance.	• On: Capacitor (i-ELOOP) capacitance status deteriorates • Off: Capacitor (i-ELOOP) capacitance status is normal
CAP_SYS 16	On/Off	Output	Displays the deterioration condition of the capacitor internal resistance.	• On: Internal resistance status in capacitor (i-ELOOP) is not normal • Off: Internal resistance status in capacitor (i-ELOOP) is normal
CAP_SYS 17	On/Off	Output	Displays the short to ground condition of the capacitor internal circuit.	• On: Capacitor internal circuit is shorted to ground • Off: Capacitor internal circuit is normal
CAP_V	V	Input	Displays the capacitor (i-ELOOP) output voltage.	• Switch ignition ON (engine off): Approx. 13.96 V • Idle: Approx. 13.78 V • Racing (engine speed above 2,000 rpm): Approx. 14.25 V
CAPA_TEMP	°C, °F	Conversion	Displays the capacitor (i-ELOOP) temperature.	• Displays the capacitor (i-ELOOP) temperature.
CHARGE_FLG	On/Off	Output	Quick charge function set condition when replacing the capacitor is displayed.	• On: Quick charge function request when replacing capacitor is sent • Off: Quick charge function request when replacing capacitor is not sent
CHG_WARN	On/Off	Output	Displays the charge warning request send status. The DC-DC convertor (i-ELOOP) sends a charge warning request to the PCM via CAN communication.	• On: Charge warning request is sent • Off: Charge warning request is not sent
CONV_OUT_V	V	Input	Displays the output voltage in voltage reduction circuit.	• Switch ignition ON (engine off): Approx. 12.54 V • Idle: Approx. 14.46 V • Racing (engine speed above 2,000 rpm): Approx. 14.48 V
CONV_STAT	Ready/Power Output(Idle)/Power Output(Normal)/ Power Output (Limit)/Safe Mode// Power Output (Stop)/Error/Default Mode	Output	Displays the voltage reduction circuit operation status.	• Ready: Preparation for voltage reduction completed • Power Output (Idle): Output stopped • Power Output (Normal): During output • Power Output (Limit): Target output value not achieved due to limit • Safe Mode: Self-protection status (voltage reduction circuit inoperable) • Power Output (Stop): When a voltage reduction circuit malfunction occurs, output from the voltage reduction circuit stops • Error: Voltage reduction circuit is malfunctioning • Default Mode: Voltage reduction circuit outputs a fixed voltage (13.5 V)
DCDC_IN_V	V	Input	Displays the DC-DC converter (i-ELOOP) input voltage.	• Switch ignition ON (engine off): Approx. 13.92 V • Idle: Approx. 14.48 V • Racing (engine speed above 2,000 rpm): Approx. 14.48 V
DCDC_OUT_A	A	Input	Displays the DC-DC converter (i-ELOOP) output current.	• Switch ignition ON (engine off): Approx. 600 mA • Idle: Approx. 400 mA • Racing (engine speed above 2,000 rpm): Approx. 400 mA
DCDC_OUT_V	V	Input	Displays the DC-DC converter (i-ELOOP) output voltage.	• Switch ignition ON (engine off): Approx. 12.54 V • Idle: Approx. 14.46 V • Racing (engine speed above 2,000 rpm): Approx. 14.46 V

Item	Unit/Condition	Type	description	Condition/Specification (Reference)
DCDC_TEMP	°C, °F	Conversion	Displays the DC-DC converter (i-ELOOP) temperature	• Displays the DC-DC converter (i-ELOOP) temperature
DEGEN_COUNT	—	Output	Displays the number of times the system switches to safe mode due to high temperature in the DC-DC converter (i-ELOOP).	• Displays the number of times the system switches to safe mode due to high temperature in the DC-DC converter (i-ELOOP).
DEGEN_WARN	Ok/Degeneration warning/ Degeneration	Output	Displays the control status due to high temperature in the DC-DC converter (i-ELOOP).	• Displays the control status due to high temperature in the DC-DC converter (i-ELOOP).
DIODE_STAT	O.K./FAULT	Output	Displays the diode circuit status.	• O.K.: Diode circuit status in DC-DC converter (i-ELOOP) is normal • FAULT: Diode circuit status in DC-DC converter (i-ELOOP) is not normal
IR_REQ	On/Off	Output	Displays i-stop restart request send status. The DC-DC convertor (i-ELOOP) sends an i-stop restart request to the PCM via CAN communication.	• On: i-stop restart request is sent • Off: i-stop restart request is not sent
IR_SW_STAT	Connect (On)/ Disconnect (Off)/ Fault (On)/Fault (Off)	Output	State of IR relay within DC-DC Converter (i-ELOOP)	• Connect (On): IR relay ON • Disconnect (Off): IR relay OFF • Fault (On): IR relay ON (Fail) • Fault (Off): IR relay OFF (Fail)
I-STOP_REQ	On/Off	Output	Displays the i-stop inhibit request send status. The DC-DC convertor (i-ELOOP) sends an i-stop inhibit request to the PCM via CAN communication.	• On: i-stop inhibit request is sent • Off: i-stop inhibit request is not sent
MAS_WARN	On/Off	Output	Displays the master warning light illumination request send status. The DC-DC convertor (i-ELOOP) sends a master warning light illumination request to the instrument cluster via CAN communication.	• On: Master warning light illumination request is sent • Off: Master warning light illumination request is not sent
OUT_M_CUR1	A	Calculation	Output Mean Current in IG On (last time)	• Displays the average value of the DC-DC converter (i-ELOOP) output current from the previous time the ignition was switched ON.
OUT_M_CUR2	A	Calculation	Output Mean Current in IG On (2 times before)	• Displays the average value of the DC-DC converter (i-ELOOP) output current from the previous 2 times that the ignition was switched ON.
OUT_M_CUR3	A	Calculation	Output Mean Current in IG On (3 times before)	• Displays the average value of the DC-DC converter (i-ELOOP) output current from the previous 3 times that the ignition was switched ON.
OUT_M_CUR4	A	Calculation	Output Mean Current in IG On (4 times before)	• Displays the average value of the DC-DC converter (i-ELOOP) output current from the previous 4 times that the ignition was switched ON.
OUT_M_CUR5	A	Calculation	Output Mean Current in IG On (5 times before)	• Displays the average value of the DC-DC converter (i-ELOOP) output current from the previous 5 times that the ignition was switched ON.

Item	Unit/Condition	Type	description	Condition/Specification (Reference)
PRECHG_WARN	On/Off	Output	Displays the pre-charge warning indication request send status. The DC-DC convertor (i-ELOOP) sends a pre-charge warning indication request to the instrument cluster via CAN communication.	• On: Pre-charge warning indication request is sent • Off: Pre-charge warning indication request is not sent
PURGE_RESLT	Charged/Purged/ Fault	Output	Result of charge purge on capacitor (i-ELOOP) after IG Off	• Displays the purge result of the capacitor (i-ELOOP) electrical charge after the ignition is switched off.